



Day-1 ML

Building a RAG System

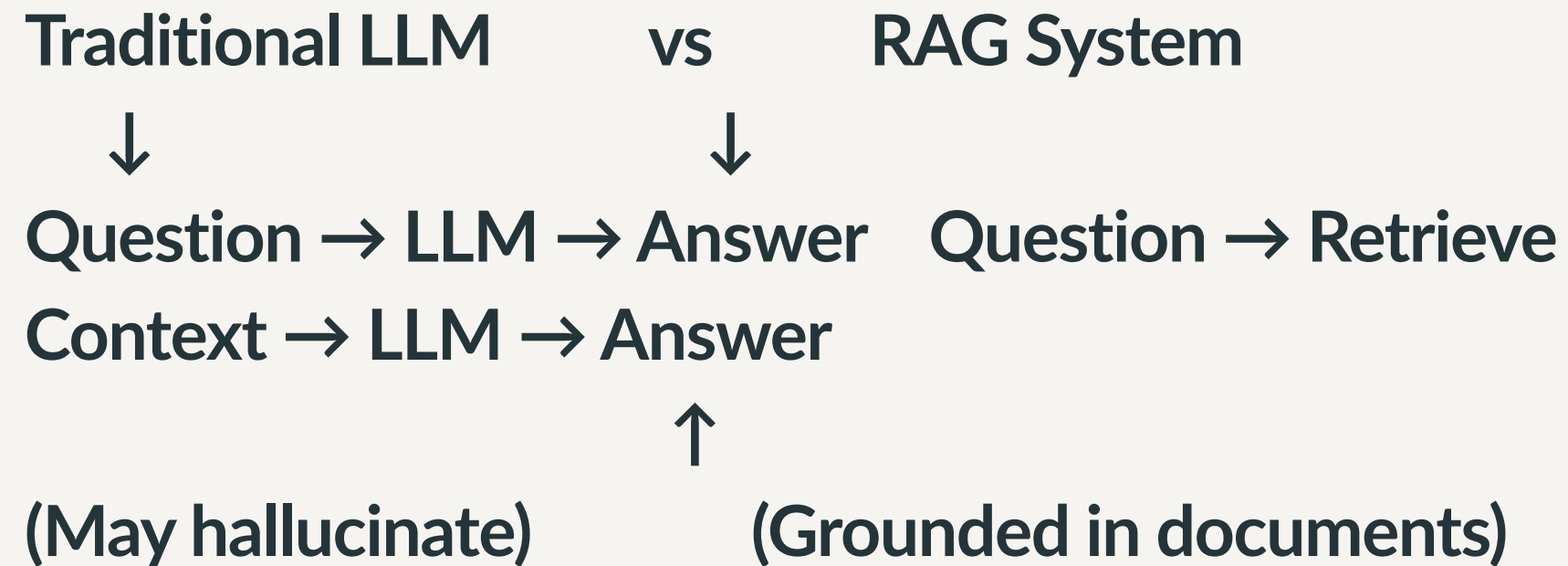


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Problem Statement

- Given: PDF document on Internal Combustion Engines (78 pages)
- Task: Build a Q&A system to answer 48 technical questions
- Goal: Answer questions using ONLY information from the PDF
- Constraints:
 - No external knowledge
 - No hallucination
 - Must cite document context
 - Fallback: "Unable to answer from provided documents"

What is RAG?



Key Benefits:

- ✓ Factual accuracy
- ✓ Source attribution
- ✓ Domain-specific knowledge
- ✓ No hallucination

System Architecture

PDF (78 pages)



Chunking (800 chars, 150 overlap)



Embeddings (sentence-transformers)



FAISS Vector Store



Question → Retrieve (Top 10) → Rerank (Top 5) → LLM → Answer

Components:

- Document Processing
- Embedding Model
- Vector Database
- Retrieval System
- LLM Generation

Technical Implementation

Component	Technology	Why?
Embeddings	sentence-transformers/all-MiniLM-L6-v2	Free, fast, 384-dim
Vector DB	FAISS	Fast similarity search
Reranker	cross-encoder/ms-marco	+10-15% accuracy
LLM	Groq (llama-3.3-70b)	High quality, fast
Framework	LangChain	RAG orchestration

Key Parameters:

- Chunk size: 800 characters
- Overlap: 150 characters
- Temperature: 0.1 (low = less creative)

Pipeline Stages - Part 1

1. EDA (Exploratory Data Analysis)

- Loaded PDF: 78 pages, ~22K characters
- Analyzed structure and content

2. Chunking

- Split into ~150 semantic chunks
- Size: 800 chars with 150 overlap
- Preserves context at boundaries

3. Embedding

- Converted chunks to 384-dim vectors
- Normalized for cosine similarity
- Created searchable representations

4. Vector Store

- Built FAISS index with ~150 vectors
- Fast similarity search (milliseconds)

5. Retrieval + Reranking

- Initial: Retrieve top-10 similar chunks
- Rerank: Cross-encoder scores relevance
- Final: Top-5 most relevant chunks

6. Answer Generation

- LLM receives question + context
- Strict prompt: "Use ONLY provided context"
- Low temperature prevents hallucination

Results & Performance

- Final Score: 0.921 / 1.0
- Score Distribution:
- ● 1.0 (Perfect): 28 answers (60%)
- ● 0.9-0.99: 15 answers (32%)
- ● 0.8-0.89: 3 answers (6%)
- ● 0.7-0.79: 1 answer (2%)
- Total Questions: 47/48 answered Success Rate: 97.9%
- Evaluation Method: LLM-as-Judge with detailed reasoning

What I Learned

- RAG Architecture
 - >Understanding retrieval-augmented generation
 - >Combining search with LLMs
- Vector Databases
 - >Embeddings and similarity search
 - >FAISS for efficient retrieval
- Prompt Engineering
 - >Preventing hallucination
 - >Context-aware prompting
- Reranking
 - >Two-stage retrieval improves accuracy
 - >Cross-encoders vs bi-encoders
- Evaluation
 - >LLM-as-Judge methodology
 - >Automated scoring systems